

Infrastructure Cost & Deployment Strategy

SN Polymers Pvt. Ltd.

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Infrastructure Cost Report for SN Polymers Pvt. Ltd.

Executive Summary

This report presents a management-level cost and deployment strategy for the proposed internal ERP application at SN Polymers Pvt. Ltd. The recommended approach is to adopt a lean initial cloud deployment, transition to a production-ready configuration when the system goes live, and scale infrastructure investment in alignment with business growth beyond the current 30-user baseline.

The core message for management is straightforward: SN Polymers can begin with a right-sized cloud infrastructure, operate reliably on fully managed services, and expand investment only as business usage warrants. This approach minimises upfront commitment, keeps ongoing operating costs predictable, and preserves full capacity to scale as the organisation grows.

One Page Financial Snapshot

Item	Cost
Initial Deployment	₹25,000
Monthly Operating Cost	₹700–₹4,500
Annual Operating Cost	₹8,400–₹54,000
Five-Year Estimate – Optimistic	₹2 lakh
Five-Year Estimate – Expected	₹3 lakh
Five-Year Estimate – Conservative	₹4 lakh

This page is intended as the quickest management summary. It shows the likely spending band from a minimum internal-use setup to a more production-ready setup, without assuming unnecessary enterprise infrastructure at the start.

Cost Options

Option A – Development / Pilot Environment

This option represents the lowest practical infrastructure investment for internal use, and is well suited to pilot deployments, controlled rollouts, or early internal testing. It assumes Vercel Hobby, Render Starter, Supabase Free, one domain, and free internal messaging channels.

Service	Plan	Monthly Cost
Frontend	Vercel Hobby	₹0
Backend	Render Starter	₹600
Database	Supabase Free	₹0
Domain	Standard domain	₹80
Telegram	Free	₹0
Email	Free SMTP	₹0
Total		≈ ₹700/month

Use case: Internal pilot, controlled employee rollout, or very low-risk early production use.

This is the most cost-efficient entry point, but it is not the optimal long-term configuration for a formal production system because the free database tier has limited capacity and fewer recovery features than the Pro tier.

Option B – Recommended Production Environment ★

This is the recommended configuration when the ERP goes live for normal internal business use. It upgrades the database to Supabase Pro and the frontend to Vercel Pro while keeping the backend on a low-cost Render plan.

Service	Monthly Cost
Vercel Pro	₹1,700

Render Starter	₹600
Supabase Pro	₹2,100
Domain	₹80
Total	≈ ₹4,500/month

Use case: Recommended if the company is going live and wants a better balance of reliability, production readiness, backups, and business continuity. This is the main recommended option for SN Polymers.

Option C – Future Growth Environment

This option is for future planning when the company expands materially, such as 100+ users, higher document volume, heavier dashboards, more transactions, or stricter reporting and compliance needs. At that stage, infrastructure cost will rise mainly because of larger backend capacity, stronger database resources, and additional support tooling.

Growth Stage	Estimated Monthly Cost
Moderate growth	₹10,000+
Strong growth	₹20,000+
Larger enterprise-style setup	₹50,000+

Management takeaway: infrastructure investment begins at the lowest practical level and scales only as business growth requires. This staged approach avoids premature overinvestment.

CAPEX vs OPEX

Non-technical management often prefers the cost structure in capital and operating terms because it fits budgeting discussions more clearly.

CAPEX — One-Time Costs

CAPEX Item	Meaning	Estimated Cost
Development	Application build effort	Excluded from this infrastructure report
Deployment	₹0 (internal effort; external vendor: ₹8,000–₹15,000)	₹0 (in-house)
Initial setup	₹0 (internal effort; external vendor: ₹10,000–₹20,000)	₹0 (in-house)
Domain purchase	₹900–₹1,800 (3-year registration)	₹900–₹1,800

OPEX — Recurring Costs

OPEX Item	Meaning	Estimated Range
Hosting	Frontend and backend cloud services	₹600–₹2,300/month
Database	Managed PostgreSQL and storage	₹0–₹2,100/month
Domain	Annual renewal	₹80/month equivalent
Maintenance	Application support and operational reserve	₹1,000+/month
Cloud services	Backups, storage growth, monitoring, logs	Variable by usage

This classification helps management separate one-time launch spending from recurring operational cost, which makes approval and budgeting much easier.

Deployment Cost

The following costs are one-time costs associated with making the ERP live and usable in a production business environment.

One-Time Item	Estimated Cost (INR)	Notes
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Domain purchase	₹900–₹1,800 (3-year registration)	Initial registration
Initial deployment	₹0 (internal effort; external vendor: ₹8,000–₹15,000)	Live setup and publishing
Configuration	₹5,000–₹10,000	Environment setup and integrations

For in-house delivery, the only out-of-pocket deployment cost is the domain registration (**₹900–₹1,800 over 3 years**). If an external vendor is engaged for deployment, configuration, and go-live support, the planning budget is approximately **₹25,000**, with a broader band of **₹22,900 to ₹46,400** depending on scope.

Annual Maintenance and Application Support

The maintenance section is better understood when split into **mandatory**, **operational**, and **optional** categories.

Mandatory

Item	Estimated Yearly Cost	Notes
Hosting	₹7,200–₹27,600	Depends on whether current or production-ready stack is used.
Domain	₹900–₹1,800 (3-year registration)	Annual renewal
Database	₹0–₹25,200	Depends on free vs Pro tier.

Operational

Item	Estimated Yearly Cost	Notes
Annual Application Support	₹15,000–₹40,000	Replaces the term “bug fixes” with a more professional support category.

Small feature requests	₹20,000–₹60,000	Minor reports, fields, and workflow adjustments
Security updates	₹5,000–₹15,000	Periodic hardening and patching

Optional

Item	Estimated Yearly Cost	Notes
SMS services	Variable	Only if SMS becomes required
New modules	Project-based	Future business expansion work

This structure makes the annual maintenance section look more professional and helps management distinguish unavoidable running cost from discretionary enhancement spending.

Infrastructure Comparison Table

Approach	Initial Cost	Monthly Cost	Recommendation
Own Server	High	High	✗ Not Recommended
VPS	Medium	Medium	△ Acceptable
Current Cloud Stack	Low	Low	✓ Recommended

The current managed cloud stack is recommended because it gives the best balance of low entry cost, lower technical maintenance burden, and easier scaling for a manufacturing company running an internal ERP.

Decision Matrix

Option	Cost	Reliability	Maintenance Effort	Recommendation
Self-host	High	Medium	High	Not Recommended
Cloud (Current)	Low	High	Low	Recommended
AWS Enterprise	Very High	Very High	Medium	Overkill

This consultant-style view helps management compare choices quickly. The current cloud approach scores best because it combines low cost, high reliability, and minimal internal support burden for the present scale.

Why Not AWS or Azure?

Large enterprise cloud platforms such as AWS and Azure offer excellent scalability and a wide range of services, but they would introduce unnecessary cost, configuration effort, and operational complexity for a 30-user internal ERP at this stage. The recommended managed cloud stack provides sufficient reliability and business continuity at a significantly lower recurring cost and with far less day-to-day administration.

Vendor Lock-In Note

Management should also be aware of **vendor lock-in** risk, which means some future migration effort may be required if the company later decides to move away from Vercel, Render, or Supabase. In this case, the risk is moderate rather than severe, because the application uses widely understood technologies such as React, Node.js, Express, and PostgreSQL, which are portable to other hosting environments if needed.

Return on Investment

Management decisions are easier when the software is framed not only as a cost, but also as a business time-saving tool.

Expected Business Benefits

- Eliminates paper-based approvals.
- Speeds up report generation.
- Centralizes document storage.
- Reduces manual entry errors.
- Improves real-time project tracking.

- Makes audits easier.
- Increases accountability across departments.

Indicative Productivity Impact

Even modest daily time savings across employees can create meaningful business value through faster approvals, better reporting, reduced manual work, and fewer process delays.

This means the ERP should be viewed not only as a software cost, but also as a productivity improvement tool that can reduce administrative effort, improve accountability, and support smoother audits and reporting.

Scaling View

The system can scale in phases rather than through a large upfront investment.

User Count	Estimated Monthly Infrastructure Cost
30 users	₹700–₹4,500
50 users	₹6,500–₹8,000
100 users	₹10,000+
250 users	₹20,000+
500 users	₹50,000+

The first areas likely to need upgrades are the backend, database resources, and storage. This is a healthy cost pattern because the company pays more only when adoption and transaction volume actually increase.

Five-Year Estimate

A conservative five-year management estimate, depending on how quickly the system matures from low-cost internal use to a more stable production setup, is approximately **₹2 lakh to ₹4 lakh**. This range assumes the company starts with a low-cost model, moves to production readiness, and adds limited maintenance and optional support over time.

Assumptions and Exclusions

To avoid misunderstandings later, the report should clearly separate what is included in this estimate and what is not.

All costs are planning estimates based on publicly available cloud pricing as of June 2026 and may vary due to exchange rates, taxes, or future pricing changes.

Included

- Hosting
- Database
- Domain
- SSL
- Storage
- Email
- Telegram notifications
- Basic backups

Excluded

- Development manpower
- Future feature development
- Third-party integrations
- Premium monitoring tools
- WhatsApp Business API, unless explicitly approved
- Disaster recovery infrastructure

These boundaries are important because they prevent management from confusing infrastructure cost with full software project cost.

Security Baseline

The proposed cloud providers include managed SSL encryption, secure data transmission, role-based access controls implemented by the application, and managed database services, providing an appropriate security baseline for an internal ERP deployment.

Final Recommendation

SN Polymers should adopt a phased cloud approach. Start with the low-cost internal setup if the objective is a controlled rollout, then move to the production-ready configuration when the ERP becomes part of routine business operations.

The most practical recommendation for management is:

1. Begin with the lowest practical infrastructure investment.
2. Go production-ready when usage stabilizes.
3. Scale infrastructure investment in line with business growth.

This approach is sufficient for the current scale, scalable for future growth, secure enough for a normal internal business system, and much more cost-effective than buying dedicated servers now.